

B.Tech Degree IV Semester Examination April 2011

CE 402 A/B SURVEYING II (2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART - A (Answer ALL questions)

(8 x 5 = 40)

- I. (a) Which are the elements of a simple curve?
(b) How will you arrive at the length of a transition curve?
(c) What are the factors that influence the selection of stations in a triangulation survey?
(d) Explain the figure adjustment of a plane triangle.
(e) Explain any one method of determination of latitude of a place.
(f) If the longitude of the place of observation, is $93^{\circ} 45' 45'' E$, find the L.M.T of the place.
(g) Explain any one method of locating soundings.
(h) Determine the flying height of the camera above M.S.L if the focal length is 150 mm, the average elevation of the ground surface is 1500 m and the scale of photograph is 1 : 10,000.

PART – B

(4 x 15 = 60)

- II. The straight lines intersect at a chainage of 1150.50, and the angle of intersection is 60° . If the radius of the curve is 500 m, determine :
- (i) tangent distance
 - (ii) length of the curve
 - (iii) chainage of points of curvature and tangency
 - (iv) length of the long chord
 - (v) degree of curve
 - (vi) apex distance and the mid – ordinate
- (15)
- OR**
- III. Two tangents intersect at the chainage 1190 m, the deflection angle being 36° . Calculate all the data necessary for setting out a curve with a radius of 300 m by deflection angle method. The peg interval is 30 m. (15)
- IV. (a) What do you understand by a satellite station? (5)
(b) Two triangulation stations A and B are 45 Km apart. Their elevations are 244.45 m and 275.00 m respectively. The intervening ground may be assumed to have a mean elevation of 222 m. Find the minimum height of the signal required to be erected at B so that the line of sight may not graze the ground less than 3 m. (10)
- OR**
- V. (a) Which are the different laws of weight? (7)

(P.T.O.)

- (b) The following observations refer to the values of the angles A, B, C at a triangulation station :

$$A = 20^{\circ} 18' 26.4''$$

$$B = 36^{\circ} 28' 14.8''$$

$$C = 56^{\circ} 46' 44.5''$$

Fulfilling the condition that $A + B = C$, find the most probable values of A, B and C.

(8)

VI.

Find the declination and the hour angle of a star from the following data :

$$\text{Latitude of the place} = 48^{\circ} 30' N$$

$$\text{Azimuth of the star} = 50^{\circ} W$$

$$\text{Altitude of the star} = 28^{\circ} 24'$$

(15)

OR

VII.

Find the L.S.T at a place in longitude $76^{\circ} 30' E$ at 4 h 30 m P.M, G.S.T at G.M.N being 4 h 36 m 18 s.

(15)

VIII.

P, Q and R are three visible stations in a hydrological survey. The computed sides of the $\triangle PQR$ are : $PQ = 1000$ m, $QR = 1300$ m and $RP = 1900$ m. Outside the $\triangle PQR$ and nearer to PR a station X is established and its position is to be found by three point resection P, Q and R, the angles PXQ and QXP being respectively 40° and 50° , find PX and RX.

(15)

OR

IX.

- (a) Explain the principle of the method of Terrestrial photogrammetry.
(b) A ground area 30 Km x 20 Km is to be covered by aerial surveying. The following data are available :

(5)

$$\text{Format size} : 230 \text{ mm} \times 230 \text{ mm}$$

$$\text{Scale} : R.F = 1 : 20,000$$

$$\text{Longitudinal overlapping} : 60\%$$

$$\text{Lateral overlapping} : 20\%$$

Calculate the air base length and also the number of photographs required.

(10)